The average velocity is also a vector quantity. But in one dimension we have only two directions (positive and negative x direction), hence we use positive and negative signs to denote the direction.

The instantaneous velocity or velocity is defined as $v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$

Graphically the slope of the position-time graph will give the velocity of the particle. At the same time, if velocity time graph is given, the distance and displacement are determined by calculating the area under the curve. This is explained below.

We know that velocity is given by $\frac{dx}{dt} = v$.

Therefore, we can write $dx = vdt$

By integrating both sides, we get $\int_{x_1}^{x_2} dx = \int_{t_1}^{t_2} v\, dt$.

As already seen, integration is equivalent to area under the given curve. So the term $\int_{t_1}^{t_2} v\, dt$ represents the area under the curve $v$ as a function of time.

Since the left hand side of the integration represents the displacement travelled by the particle from time $t_1$ to $t_2$, the area under the velocity time graph will give the displacement of the particle. If the area is negative, it means that displacement is negative, so the particle has travelled in the negative direction. This is shown in the Figure 2.35 below.

## EXAMPLE 2.25

A particle moves along the *x-axis* in such a way that its coordinates $x$ varies with time 't' according to the equation $x = 2 - 5t + 6t^2$. What is the initial velocity of the particle?

### *Solution*

$$x = 2 - 5t + 6t^2$$

$$\text{Velocity, } v = \frac{dx}{dt} = \frac{d}{dt}(2 - 5t + 6t^2)$$

$$\text{or } v = -5 + 12t$$

For initial velocity, $t = 0$

$\therefore$ Initial velocity $= -5 \text{ms}^{-1}$

The negative sign implies that at t = 0 the velocity of the particle is along negative x direction.

Average speed = total path length / total time period

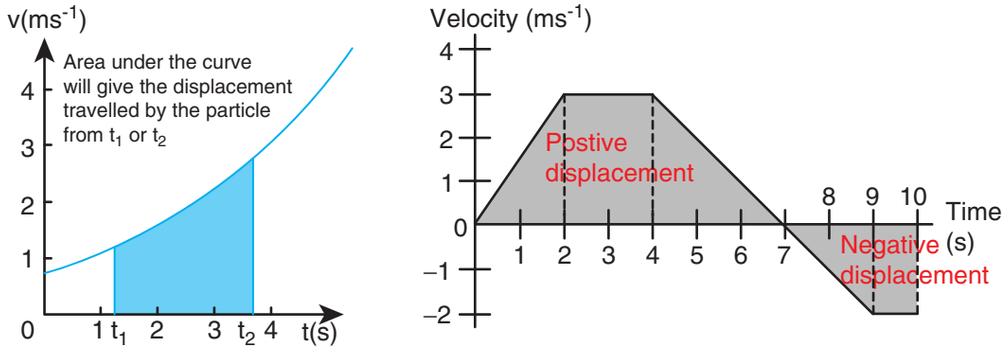



**Figure 2.35** Displacement in the velocity–time graph



UNIT-2(XI-Physics_Vol-1)-1.indd 69 05-01-2022 22:15:04